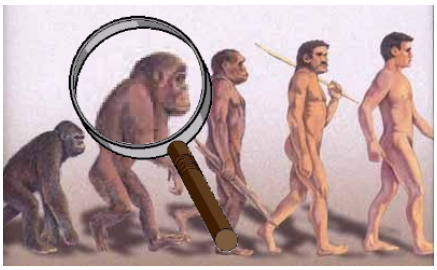
Computational Biology

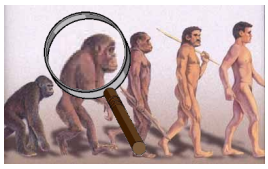
Lecture 12: *Stochastic Differential Equations*



Fisher-Wright model, SDEs, Fokker-Planck.

Outline

1. **Recap**
2. Diffusion Models
3. Fokker-Planck Equation
4. Steady State Distribution



Recap

- Recall the Fisher-Wright model looks at how a mutant allele at a single locus would take over the population
- For finite populations this is a stochastic process
- We can simulate this happening (each simulation will give us a different result)
- We can also compute $\boldsymbol{p}(t) = \big(p_0(t), p_1(t), \ldots, p_P(t)\big)$ telling us the probability of n mutants at time t using a Markov chain analysis
- Unfortunately, we can only do this for small populations

Stochasticity in Biology

- By its very nature dynamics in biology is very random
- Sometimes this doesn't matter (we can ignore the randomness)
- But sometimes this randomness is an essential part of nature
- This is particularly true when the dynamics depends on rare events that might not happen
- This is true in population genetics, but also in gene regulation which we explore later in the course

Language of Random Dynamics

- When the underlying dynamics involve small random kicks then we can develop a calculus known as **stochastic differential equation**
- This is used across the sciences
- It was originally used to describe physical diffusion
- It is used, for example, in describing fluctuations in finance

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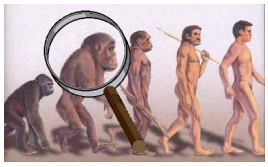


Diffusion Models

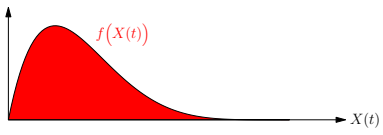
- Markov analysis quickly becomes intractable
- For weak selection and small mutation rates we can model the system using a *stochastic difference equation*
- Let $X(t)$ be the proportion of mutants at time t
- We make the assumption that at time $t + \Delta t$ the proportion of mutants is given by

$$X(t + \Delta t) = X(t) + a(X(t))\Delta t + b(X(t))\Delta W(t)$$

- where $\Delta W(t)$ is a random variable with mean 0 and variance Δt

Random Variables

- $X(t)$ (the proportion of mutants at time t) is a random variable
- That is, it varies from realisation to realisation
- $X(t)$ will be distributed according to a probability density $f(X(t))$



- We denote its average or **expectation** as

$$\mathbb{E}\left[X(t)\right] = \int X(t)\, f(X(t))\, \mathrm{d}t = \langle X(t) \rangle$$

Naming and Mis-naming

- On average

$$\mathbb{E}\left[X(t+\Delta t)\right] = X(t) + a(X(t))\Delta t$$

this defines the function $a(X(t))$

- Physicists call this the *drift term*
- The variance in $X(t+\Delta t)$ is given by

$$\mathbb{E}\left[X^2(t+\Delta t)\right] - \mathbb{E}\left[X(t+\Delta t)\right]^2 = b^2(X(t))\Delta t$$

this defines the function $b(X(t))$

- Population geneticists call $b(X(t))\Delta W(t)$ *genetic drift*

Fisher-Wright Average

- For the Fisher-Wright model we have

$$\mathbb{P}\left(X(t+\Delta t) = x\right) = \mathbb{P}\left(n' = Px\right) = \binom{P}{Px} p_{sm}^{Px}(1-p_{sm})^{P(1-x)}$$

- For this model it makes sense to take $\Delta t = 1$ (one generation)
- Taking averages

$$a(X(t)) = \mathbb{E}\left[X(t+1)\right] - X(t) = p_{sm} - X(t)$$
$$= \frac{(u - (u+v-s+u\,s)X(t) + sX^2(t))}{1+s\,X(t)}$$

Fisher-Wright Variance

- The variance of a binomial is given by

$$b^2(X(t)) = \mathbb{E}\left[X^2(t+1)\right] - \mathbb{E}\left[X(t+1)\right]^2 = \frac{p_{sm}(1-p_{sm})}{P}$$

- If s, u and v are small then $p_{sm} \approx X(t)$ so

$$b^2(X(t)) \approx \frac{X(t)\left(1-X(t)\right)}{P}$$

- Recall

$$X(t+\Delta t) = X(t) + a(X(t))\Delta t + b(X(t))\Delta W(t)$$

Continuous Time

- Taking the limit $\Delta t \to 0$ we obtain a (Itô) *stochastic differential equation* (SDE)

$$\mathrm{d}X(t) = a(X(t))\,\mathrm{d}t + b(X(t))\,\mathrm{d}W(t)$$

- This has to be viewed as a formal definition—its precise meaning is rather technical
- When we treat $X(t)$ as a function of t then the function is known as a **stochastic process**

Why SDE Work

- SDE's model fluctuating systems using two terms
  - The mean change described by $a(X(t))\,\mathrm{d}t$ which we can model accurately (it become more accurate in large populations where the difference between discrete and continuous time is negligible)
  - Fluctuations $b(X(t))\,\mathrm{d}W(t)$ which are assumed to be normally distributed
- Although our fluctuations are not normally distributed the accumulative effect of many fluctuations will be very close to normally distributed due to the central limit theorem

Recap

- There are a lot of techniques for dealing with *stochastic differential equation*

$$\mathrm{d}X(t) = a(X(t))\,\mathrm{d}t + b(X(t))\,\mathrm{d}W(t)$$

- The Fisher-Wright model is well approximated by an SDE with

$$a(X(t)) = \frac{(u - (u+v-s+u\,s)X(t) + sX^2(t))}{1+s\,X(t)}$$
$$b^2(X(t)) = \frac{X(t)\left(1-X(t)\right)}{P}$$

- Unfortunately it is too difficult to obtain a direct solution of the form $X(t) = g(W(t))$

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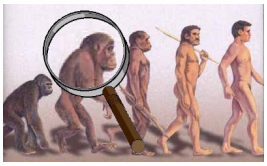


Random Instances and Distributions

- The stochastic differential equation (aka diffusion equation) describes $X(t)$ in terms of a second stochastic process $W(t)$
- $W(t)$ is known as a Weiner measure it has the distribution

$$f\left(W(\Delta t + t_0) - W(t_0) = x\right) = \mathcal{N}(x|0, \Delta t)$$

- where $\mathcal{N}(x|\mu, \sigma^2)$ is a normal (Gaussian) distribution with mean μ and variance σ^2
- However $W(t)$ varies from instance to instance so even if we can solve the stochastic differential equation the solution is stochastic

Fokker-Planck Equation

- Although we cannot obtain a closed form solution $X(t) = g(W(t))$, we can obtain a solution for the the probability distribution for $X(t)$

$$f(x,t)\,\mathrm{d}x = \mathbb{P}\left(x < X(t) < x + \mathrm{d}x\right)$$

- The dynamics of $f(x,t)$ going forward in time is given by the **Fokker-Planck** or **forward Kolmogorov** equation

$$\frac{\partial f(x,t)}{\partial t} = -\frac{\partial\left(a(x)\,f(x,t)\right)}{\partial x} + \frac{1}{2}\frac{\partial^2\left(b^2(x)\,f(x,t)\right)}{\partial x^2}$$

Going Back

- Diffusion isn't reversible so the equations for going back in time are different—the **backward Kolmogorov** equation

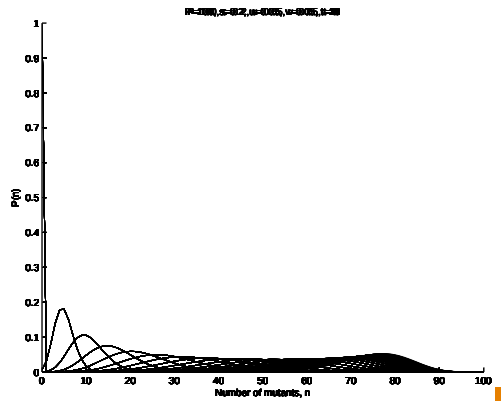
$$\frac{\partial f(x,-t)}{\partial t} = -a(x)\frac{\partial f(x,-t)}{\partial x} - \frac{1}{2}b^2(x)\frac{\partial^2 f(x,-t)}{\partial x^2}$$

- The backward equation is very useful to study systems with *absorbing states*
- It can be used to calculate the probability of finishing in an absorbing state and the time taken to reach that state

Using Fokker-Planck

- The Fokker-Planck equation helps in that rather than work with the weird world of stochastic differential equation we can work with the well understood partial differential equations (PDEs)
- The solution describes how the evolution of a PDE evolves over time
- Unfortunately it is often hard to solve PDEs in closed form (they usually involve series expansions in nasty polynomials)
- We often have to solve them numerically

Distributions Evolving Over Time



Solution to Fokker-Planck

- I cheated, this is actually the solution to the Markov chain
- However, the Fokker-Planck provides the same information and would provide an almost identical outcome
- The Markov chain solution is exact (it does not make the assumption of continuous time or that $X(t)$ is a continuous variable)
- But the Fokker-Planck equation provides a near identical solution, but is solvable for any P (Markov chain analysis becomes too slow to solve for large P)

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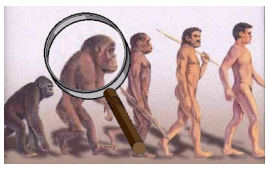


Steady State

- Although solving PDEs are in general hard
- The problem simplifies if we consider the *steady state* given by

$$\frac{\partial f(x,t)}{\partial t} = 0$$

- This (usually) happens when $t \to \infty$
- Our Fokker-Planck equation then becomes

$$\frac{\partial f(x,t)}{\partial t} = -\frac{\partial\left(a(x)\,f(x,t)\right)}{\partial x} + \frac{1}{2}\frac{\partial^2\left(b^2(x)\,f(x,t)\right)}{\partial x^2} = 0$$

Steady State Solution

- The steady state solution $f_{eq}(x) = \lim_{t\to\infty} f(x,t)$ does not depend on t so the Fokker-Planck equation becomes an ordinary differential equation

$$-\frac{\mathrm{d}\left(a(x)\,f_{eq}(x)\right)}{\mathrm{d}x} + \frac{1}{2}\frac{\mathrm{d}^2\left(b^2(x)\,f_{eq}(x)\right)}{\mathrm{d}x^2} = 0$$

- Which has a solution

$$f_{eq}(x) \propto \frac{1}{b^2(x)}\,\mathrm{e}^{\,2\int_{1/2}^{x}\frac{a(y)}{b^2(y)}\mathrm{d}y}$$

Steady State for Fisher-Wright

- Assuming s, u and v are small, then expanding to first order

$$a(X) = \mathbb{E}\left[\Delta X\right]/\Delta t \approx -v\,X + u(1-X) + sX(1-X)$$
$$b(X) = \left(\mathbb{E}\left[(\Delta X)^2\right] - \mathbb{E}\left[\Delta X\right]^2\right)/\Delta t \approx \frac{X(1-X)}{P}$$

- Using these values we find

$$f_{eq}(x) = \frac{x^{2\,P\,u-1}\,(1-x)^{2\,P\,v-1}\,\mathrm{e}^{2\,P\,s\,x}}{B(2\,P\,u, 2\,P\,v)\,F(2\,P\,u; 2\,P(u+v); 2\,P\,s)}$$

- The denominator involves special function, but is just a normalisation constant

What's the Point

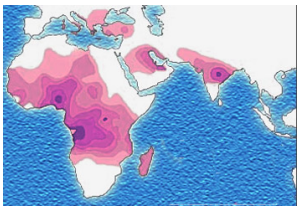
- This seems a long journey to get a complicate formula, but this tells us the final distribution as a function of the parameters of our model
- As evolution is slow the steady-state solution is often what we observe
- With more work we can also obtain an understanding of the dynamics (e.g. typical take over time)
- One very important observation is that the take over times are fast even when the mutants have only a small advantage

Sickle-Cell Anaemia

- One classic example of population genetics is to understand the prevalence of sickle-cell anaemia
- This is caused by a particular mutation, but carriers only suffer if they have two copies of the mutation
- Nevertheless, sickle-cell anaemia is very serious and is disadvantageous enough that it should be selected out of the population
- Except, when a carrier has a single copy it helps fight against malaria and so provides a selective advantage

Sickle-Cell Prevalence

- Thus, in malaria regions there is a balance between the fitness gain from helping to fight malaria and the fitness loss due to the offspring that inherit two copies of the mutation



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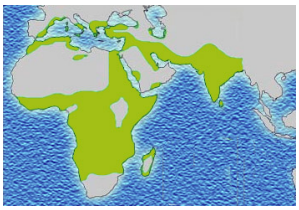

Historic Malaria Prevalence

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Conclusion

- Modelling the dynamics of evolving populations has a long distinguished history
- Finite population effects are crucial to understanding evolution
- With modern computers we can now simulate evolutionary models very easily
- Solving mathematical models forces you to understand the mechanisms that drive evolutionary dynamics
- Stochastic differential equations are ubiquitous in modelling a lot of biological problems